

Shubham Saluja Kumar Agarwal

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EDUCATION

Cornell University | Ithaca, NY

August 2026 – May 2027

Master of Engineering in Electrical and Computer Engineering

Purdue University | West Lafayette, IN

September 2022 – May 2026

Bachelor of Science in Computer Engineering

Concentrations: Computer Systems; Microelectronics and Semiconductors

Minors: Mathematics; Business Economics

GPA: 4.00/4.00

RESEARCH EXPERIENCE

Versatile Electronic Systems Lab | Cornell University

September 2026 – Present

Supervisor: Professor Amal El-Ghazaly

Magnetically Actuated Haptic Displays

- Designing new form factors of a soft electro-permanent-magnet actuator: fabricated a custom magnetic composite, wound the coils, and designed the enclosures.
- Running COMSOL simulations to select actuator design parameters.
- Exploring applications of the actuator platform beyond haptics.

Hybrid Body Lab | Cornell University

August 2026 – Present

Supervisor: Professor Cindy Hsin-Liu Kao

EdemaFlex – Hand edema treatment glove

- Fabricating knit gloves with shape memory alloy (SMA) springs for hand-edema treatment, including spring placement, wiring and functional testing.
- Extending the glove's web interface to support user-data upload, in-browser firmware flashing, and testing of individual SMA springs.

Responsive Environments Group | Massachusetts Institute of Technology

May 2026 – Present

Supervisor: Professor Joseph A. Paradiso

Miniaturized Fiber-Embedded Technologies

- Designing circuits for a further-miniaturized system-in-fiber platform.
- Responsible for component selection, schematic design, PCB layout and firmware.

Nanomaterials Lab | Universidad Autónoma del Estado de Morelos

May 2026 – August 2026

Supervisor: Prof. Vivechana Agarwal

Machine Learning applied to limited materials science datasets

- Trained a time-agnostic machine learning model for small-sample materials datasets to classify contaminants with an accuracy of 97% and predict concentration with a mean average error of 0.061.

QuMETA NanoML Group | Purdue University

March 2025 – Present

Supervisor: Prof. Alexander V. Kildishev, Prof. Alexandra Boltasseva, Prof. Vladimir Shalaev

Inverse Design of Photonic Devices

- Curated and validated a novel metagrating dataset of 10,000+ device structures, with the goal of open sourcing the data to accelerate community research in inverse design.
- Established performance benchmarks for a novel inverse-design ML framework by training 4 distinct baseline models on two custom datasets. Manuscript in progress.
- Implemented a VGGNet-based neural network to achieve high-fidelity predictions and eliminating the need for electromagnetic solvers, reducing efficiency evaluation time by over 100 times with minimal loss in accuracy.

Wireless Sensing Circuits and Systems Laboratory | Purdue University

April 2023 – May 2026

Supervisor: Prof. Byunghoo Jung

AirVolt – Wireless Voltage Sensing

August 2025 – May 2026

- Leading early-stage development of a wireless HVAC voltage-sensing system for use in power plants and power rail installation and maintenance.
- Outlined circuit architecture and designed initial PCB prototypes in Altium Designer for proof-of-concept testing.

Wireless Battery State of Charge Estimation

April 2023 – August 2025

- Developed a method for wireless online LiPo battery charge detection achieving relaxation-induced error under $\pm 1\%$, superior to existing state-of-the-art online sensing methods.
- Implemented an STM32-based frequency counter (30 MHz range, ± 3 Hz error) with a Python GUI for real-time sensor data

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visualization and logging.

- Created acquisition pipelines for multi-instrument battery datasets, applying filtering, noise removal, and synchronization techniques to extract reliable parameters and validate sensor/device performance.

Vertically Integrated Projects Computer Vision for Embedded Systems | Purdue University

August 2023 – May 2024

Supervisor: Prof. Yung-Hsiang Lu

Coordinating the 2024 Low Power Computer Vision Challenge (LPCVC)

- Developed a 3D reconstruction evaluation pipeline for the 2024 Low Power Computer Vision Challenge (LPCVC).
- Benchmarked open-source models to define power and accuracy performance baselines for competitor submissions.

PUBLICATIONS AND PRESENTATIONS

- **Kumar Agarwal, S. S., Iyer, V., et al.** *Benchmarking of Generative Models for Photonic Device Design*. Manuscript in Progress.
- **Kumar Agarwal, S. S., Avalos, F. et al.** *Time Agnostic Machine Learning Model for Contaminant Classification and Contaminant Concentration Prediction*. Manuscript in Progress.
- Mirani, A., **Kumar Agarwal, S. S.** (presenter), et al. *Transformer for Video Dehazing*. Poster presented at *Purdue 2024 Spring Research Talk & Poster Conference*.
- **Kumar Agarwal, S. S.** (presenter), Vidisha Singhal, et al. *Low-Power Computer Vision Challenge*. Research Talk presented at *Purdue 2023 Fall Research Talk & Poster Conference*.
- **Kumar Agarwal, S. S.,** Mendoza Enriquez, B. U., & Hernández Rivera, D. (2022). *Device Fabrication from Recycled Electronic Spare Parts: Dip Coating Device and High Voltage Power Supply Adapted for Electrospinning Device*. *Journal of Physics: Conference Series*, 2261, 012001. ICIEE 2022. DOI: 10.1088/1742-6596/2261/1/012001. Presented at the 2022 11th *International Conference on Information and Electronics Engineering (ICIEE)*

TEACHING EXPERIENCE

Bechtel Innovation and Design Center Peer Mentor

November 2022 – Present

- Directed 300+ student electronics projects from design to manufacturing, guiding diverse teams and people through PCB design, circuit prototyping, debugging, and microcontroller integration.
- Taught and mentored peers in topics ranging from basic soldering and wiring to advanced circuit troubleshooting and embedded programming, adapting lessons to varying skill levels.
- Tackled diverse technical issues by self-learning unfamiliar systems and integrating new knowledge into project solutions.

TECHNICAL SKILLS AND AWARDS

- Circuit Design and Prototyping (advanced): schematic capture, PCB layout, SMD and through-hole assembly, board bring-up and debugging (Altium Designer, KiCad, LTSpice)
- Embedded and Digital Systems (proficient): Embedded C (STM32), SystemVerilog, RISC-V, Cadence Virtuoso
- Fabrication (proficient): 3D printing, PCB milling and engraving, mechanical CAD (Autodesk Fusion 360)
- Control Systems (familiar): closed-loop control of haptic devices, current graduate coursework
- Simulation (proficient): COMSOL Multiphysics, LightTools
- Programming and Machine Learning (advanced): Python (NumPy, Pandas, PyTorch, TensorFlow), C, MATLAB, SQL
- Web (proficient): HTML, CSS, JavaScript
- Awards and Scholarships:
 - Lester B. Knight Scholar for the 2026-2027 Cornell Academic Year
 - Best Electrical and Computer Engineering Senior Design Project at the 2026 Purdue Spark Challenge
 - J. Bruce Majerus Electrical and Computer Engineering Student Scholar Award 2025
 - Honorable Mention for Electrical and Computer Engineering Undergraduate Research Awards 2024
 - Purdue Engineering Student Council Gold Industrial Roundtable Scholarship 2023
 - Best Presenter at 2022 11th *International Conference on Information and Electronics Engineering*.
- Languages: English, Spanish (native); Hindi (conversational); French (basic)